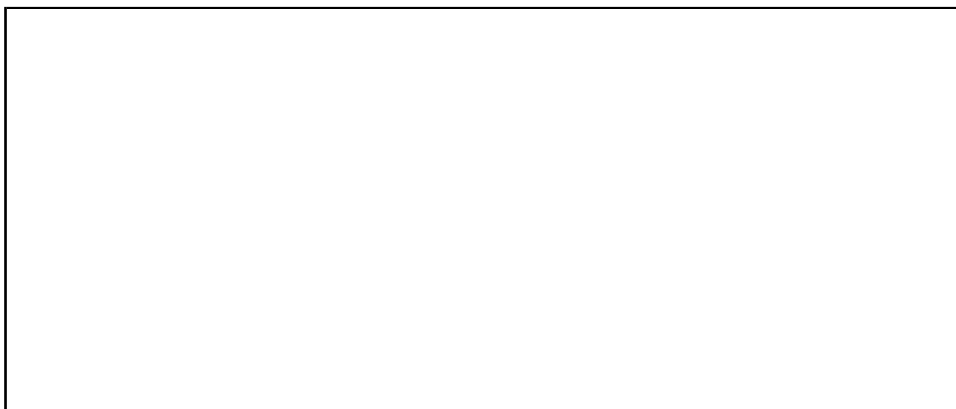


CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of M.Sc.(Physics) Theory Examination April-May 2019
PS 717 Mathematical Methods in Physics-II

Date: 01/05/2019, Wednesday **Time:** 10:00 a.m. to 10:20 a.m. **Maximum Marks:** 10

Multiple choice questions



Important Instructions:

- Provide all necessary information as required in the answer sheet.
 - Attempt all questions.
 - Answer the questions in the question paper itself.
 - Use of cell phones is strictly prohibited
 - Use of non-programmable calculators may be allowed if needed.
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Q.1. Choose the correct answer for the following questions. [10]

1. For a complex function $f(z) = |z|^2$, which of the following is true?
 - a. f is an entire function
 - b. f is analytic only at origin
 - c. f is nowhere analytic
 - d. f is not differentiable at origin.
2. $\int_C z^4 dz = \underline{\hspace{2cm}}$, where C is the boundary of a triangle with vertices $0, 1+i$ and $-1+i$.
 - a. 0
 - b. $2\pi i$
 - c. $4\pi i$
 - d. $6\pi i$.

3. The function $f(z) = \sin\left(\frac{1}{z-7}\right)$ has _____ at $z = 7$.
- simple pole
 - double pole
 - removable singularity
 - essential singularity.
4. A mapping $w = kz$, where $w = u + iv, z = x + iy, k \in \mathbb{C}$ and u, v, x, y are real variables is called _____.
- inversion
 - magnification
 - rotation
 - translation.
5. $\int_0^{\infty} x^{-\frac{1}{2}} e^{-x} dx =$ _____.
- $-2\sqrt{\pi}$
 - $2\sqrt{\pi}$
 - $\frac{\sqrt{\pi}}{2}$
 - $\sqrt{\pi}$.
6. For Bessel function $J_n(x)$, which of the following is an odd function?
- $\frac{J_3(x) - J_1(x)}{J_5(x)}$
 - $\frac{J_2(x) - J_4(x)}{J_8(x)}$
 - $\frac{J_7(x)}{J_8(x)}$
 - $\frac{J_{100}(x)}{J_{88}(x)}$.
7. For Hermite polynomials $H_n(x)$, $\int_{-\infty}^{\infty} e^{-x^2} H_7(x) H_9(x) dx =$ _____.
- $\sqrt{\pi} 2^7 7!$
 - $\sqrt{\pi} 2^9 9!$
 - $\frac{2}{15}$
 - 0.
8. For Laguerre polynomials $L_n(x)$, $L_0(500) =$ _____.
- 0
 - 1
 - 499
 - 500.
9. Which of the following is a group?
- $(\mathbb{N}, +)$
 - $(\mathbb{Z}, \cdot)$, where $\cdot$ is a multiplication
 - $(\mathbb{R}, +)$
 - $(G, \cdot)$, where $G = \left\{ \begin{bmatrix} a & b \\ d & c \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$ and $\cdot$ is a matrix multiplication.
10. A scalar is a tensor of rank _____.
- 0
 - 1
 - 2
 - 3.

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Date: 01/05/2019, Wednesday **Time:** 10:20 a.m. to 12:00 noon. **Maximum Marks:** 25

Instructions:

1. Section I and II must be attempted in separate answer sheets.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
 5. Figures to right indicate marks.
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SECTION-I

Q-2(a) Using Cauchy's integral formula, evaluate the integral $\int_C \frac{z}{(z-2)(z-1)} dz$, where [03]
 C is the circle $|z-2| = \frac{1}{2}$.

OR

(a) Find the image of $|z-3i| = 3$ under the mapping $w = \frac{1}{z}$. [03]

(b) State generating function for Legendre polynomials $P_n(x)$. Prove the differential recurrence relation $xP'_n(x) - P'_{n-1}(x) = nP_n(x)$. [03]

OR

(b) State Rodrigues formula for Hermite polynomials $H_n(x)$. Hence find $H_1(x)$ and $H_2(x)$. [03]

(c) Define: tensor and contravariant tensor. [02]

OR

(c) Let x^i and $\bar{x}^i$ be independent coordinates of a point. Show that $\frac{\partial x^j}{\partial \bar{x}^k} \cdot \frac{\partial \bar{x}^k}{\partial x^i} = \delta_i^j$. [02]

(d) Express a function $f(x) = 3x^2 - 5x + 19$ in terms of Legendre's polynomials. [02]

SECTION II

Q-3(a) Expand $f(z) = \frac{z}{(z-5)(z-3)}$ in Laurent's series valid for (i) $|z| < 3$, [03]
(ii) $3 < |z| < 5$ and (iii) $|z| > 5$.

(b) Define group. Show that a set $\{1, -1, i, -i\}$ forms an abelian group with [04]
multiplication.

OR

(b) Discuss the group of symmetries of a square. [04]

(c) For a Bessel function $J_p(x)$, show that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$. [04]

OR

(c) State Rodrigues formula for Laguerre polynomials $L_n(x)$. Show that [04]

$$L_n(x) = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!}.$$

(d) If $f = \phi + i\psi$ represents the complex potential for an electric field and [04]
 $\phi = x^3 - 3xy^2 + 3x^2 - 3y^2$, determine the function f .

OR

(d) Using Cauchy residue theorem, evaluate the integral $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$, where [04]
 $C : |z| = 3$.

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